

CSE 350

Digital Electronics and Pulse Techniques

Assignment-01

Total – 20 marks

Question 1

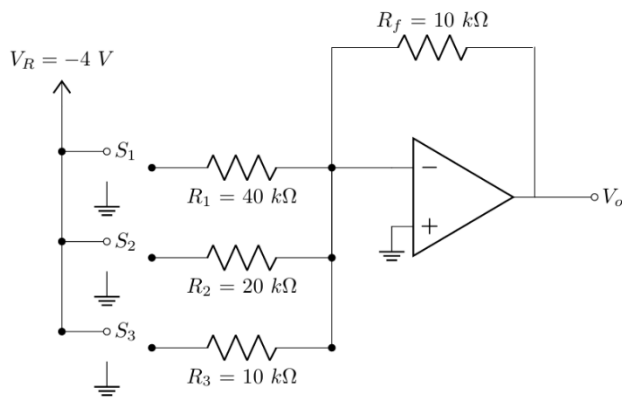


Figure Binary weighted DAC

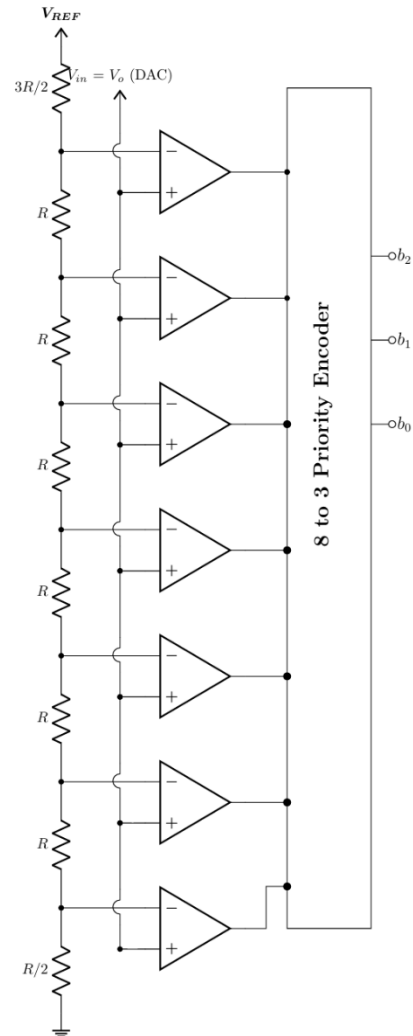
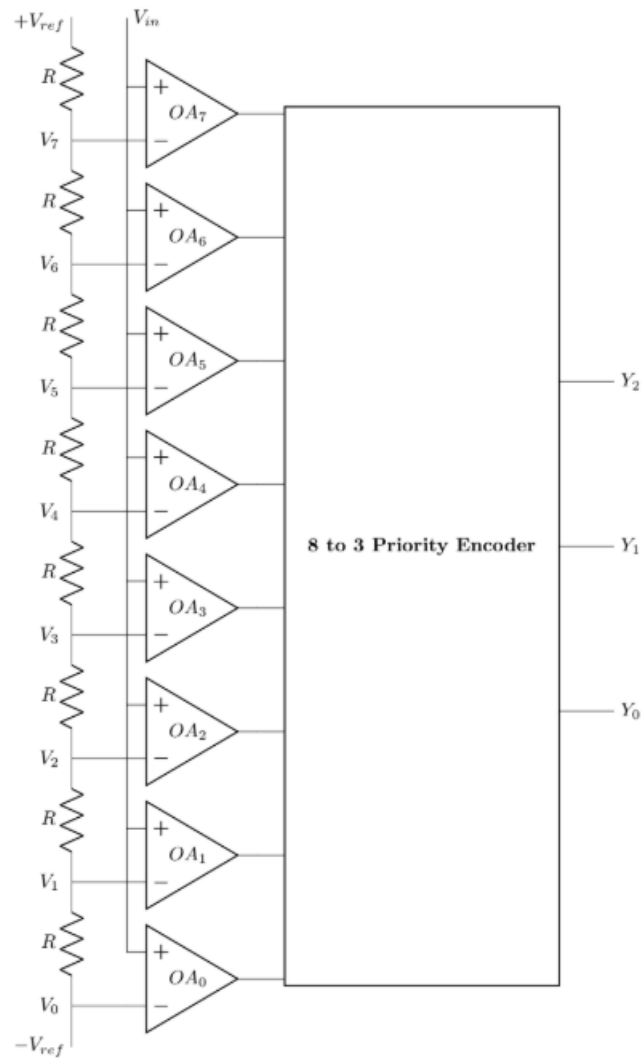


Figure Flash ADC

(a)	Determine the value of V_o of the DAC for input combinations 001 and 010.	1
(b)	Suppose the Flash ADC in Figure has an output of 010. Find the range of input voltage V_{in} that produces this output. $V_{ref} = 16$ (or any other voltage of your choice).	2
(c)	Determine all possible input combinations for the DAC that will generate V_{in} (V_o of DAC) within this range (range from b).	2

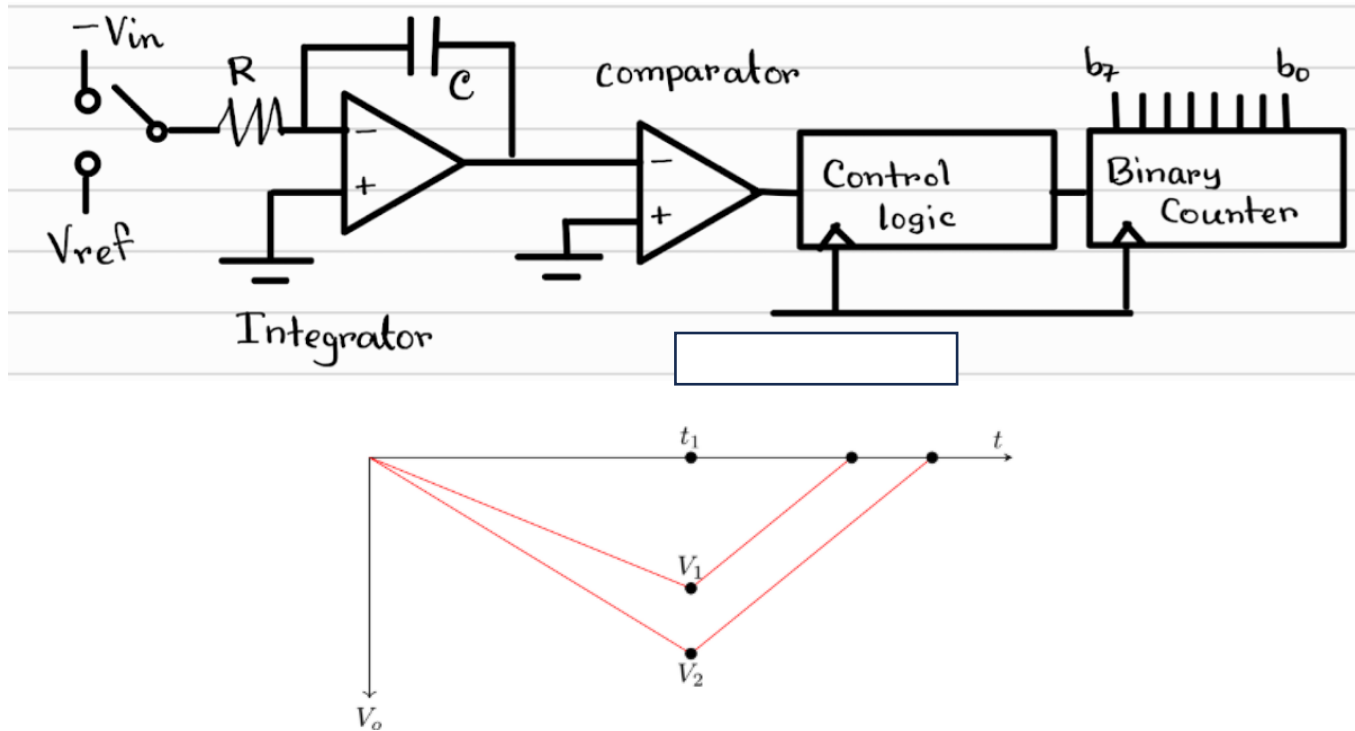
Question 2



For the given Flash ADC circuit in Figure, $V_3 = -4$ V.

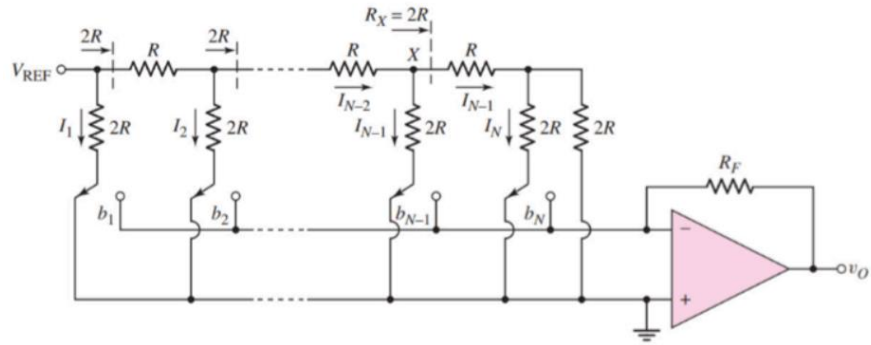
(a)	Find the reference voltages and draw the digital output vs input graph.	2.5
(b)	Now consider a 3-bit Binary Weighted DAC with the same value of reference voltage as in (a). If its maximum output voltage matches the highest quantization level of the Flash ADC, and the feedback resistance, $R_f = 2$ k Ω , draw the DAC and find the other resistances.	2.5

Question 3



a)	The integrator output voltage of a dual slope ADC $V_1 = -6\text{ V}$ and $V_2 = -8\text{ V}$. $RC = 5 \times 10^{-6}$. What are the V_{in} (Injected Sample) for these outputs? The maximum allowable sampling frequency is 1.5625 kHz .	1
b)	If the reference voltage is 15 V , find the digital outputs for the inputs found in (a) (assume an 8-bit ADC)	1
c)	Find the encoded output for which the slope of the integration phase of the integrator would be the average of the two given in a.	1
d)	If the ADC gives a digital output of 11001010 , what is the input and integrator voltage at the end of integration?	1
e)	Suppose the ADC produces a count of 153 (assume using an 8-bit ADC). What is the integrator voltage, and what is the corresponding analog input range?	1

Question 4



The N-bit DAC with an R-2R ladder network in the above figure is to be designed as a 3-bit DAC device. Suppose $V_{REF} = -10V$, $R_F = 10\text{ K}\Omega$ and $R = 5\text{ K}\Omega$.

(a)	Determine the currents I_1 , I_2 , and I_3 .	1
(b)	Determine the resolution of the circuit.	1
(c)	Determine the maximum and minimum value of V_O .	1
(d)	Determine the $\frac{1}{2}$ MSB value.	1
(e)	Draw the V_{in} Vs. V_{out} graph with proper labeling.	1